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(54) DYNAMIC SCALING OF A DISTRIBUTED COMPUTING SYSTEM

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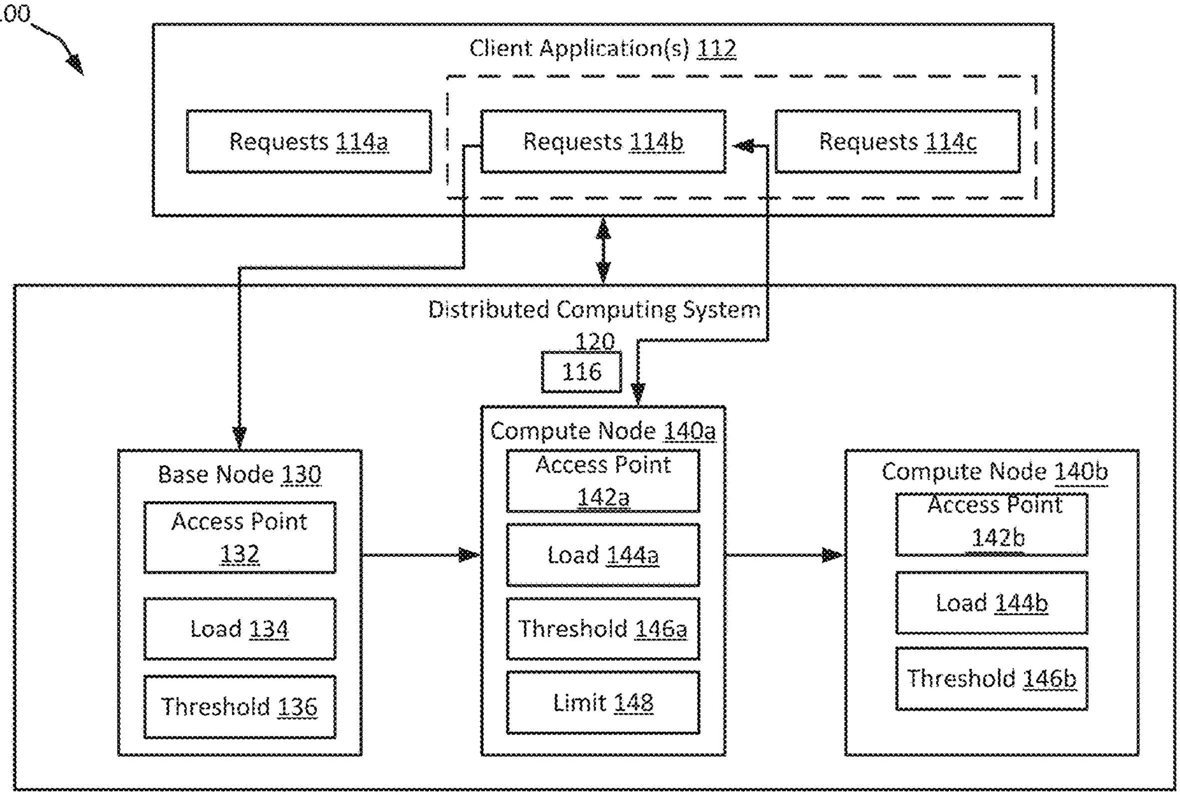
CPC *H04L 67/10* (2013.01); *H04L 67/52* (2022.05)

(57) ABSTRACT

A technique for dynamic scaling of a distributed computing system is described. In one example of the present disclosure, a system can include a base node configured to provide an access point to a distributed computing system and for servicing a first portion of requests and to generate at least one compute node based on a first load of the base node. The system can also include the at least one compute node of the distributed computing system for servicing a second portion of requests. The at least one compute node can be configured to generate an additional compute node for servicing a subset of the second portion of requests based on a second load of the at least one compute node.

100



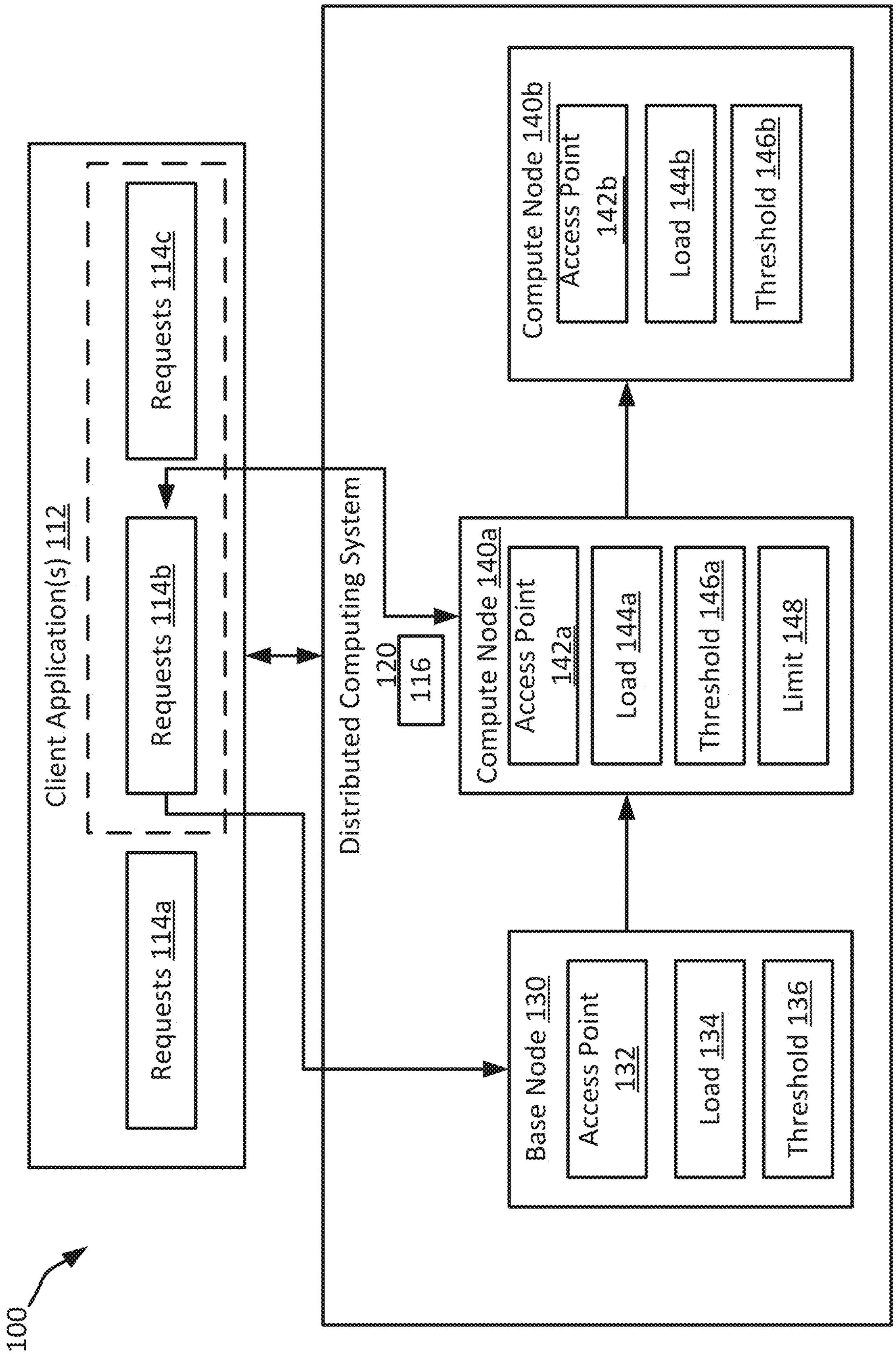

100
Client Application(s) 112
Requests 114a
Requests 114b
Requests 114c
Distributed Computing System 120
116
Base Node 130
Access Point 132
Load 134
Threshold 136
Compute Node 140a
Access Point 142a
Load 144a
Threshold 146a
Limit 148
Compute Node 140b
Access Point 142b
Load 144b
Threshold 146b

FIG. 1

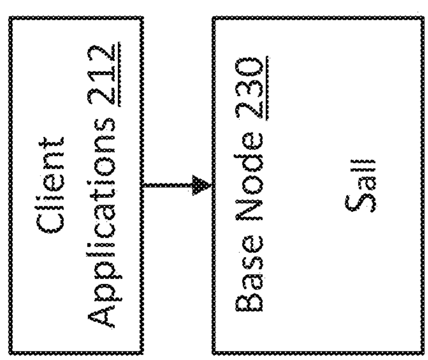


FIG. 2

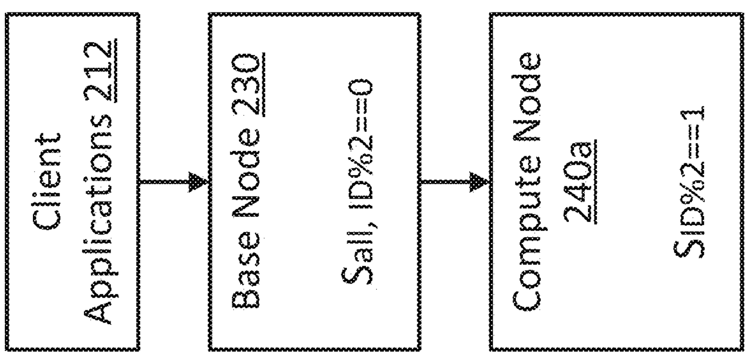

Client Applications 212
Base Node 230
S_{all}, ID%2==0
Compute Node 240a
$S_{ID\%2==1}$

FIG. 3

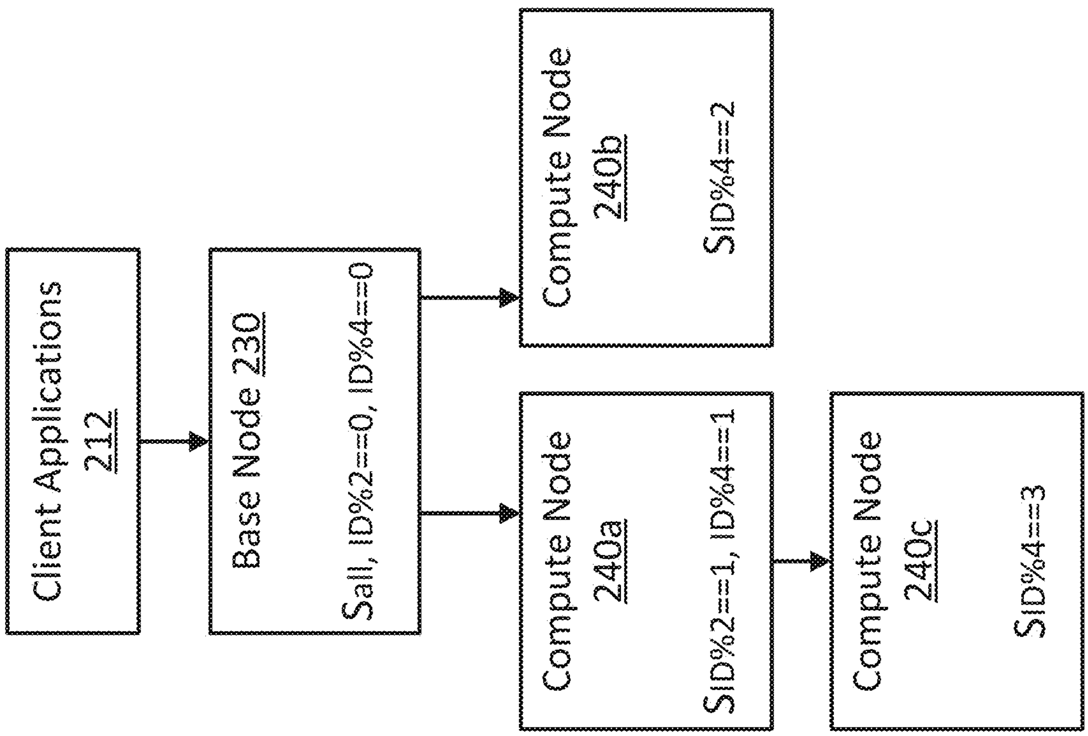

Client Applications 212
Base Node 230
S_{all}, ID%2==0, ID%4==0
Compute Node 240b
$S_{ID\%4==2}$
Compute Node 240a
$S_{ID\%2==1, ID\%4==1}$
Compute Node 240c
$S_{ID\%4==3}$

FIG. 4

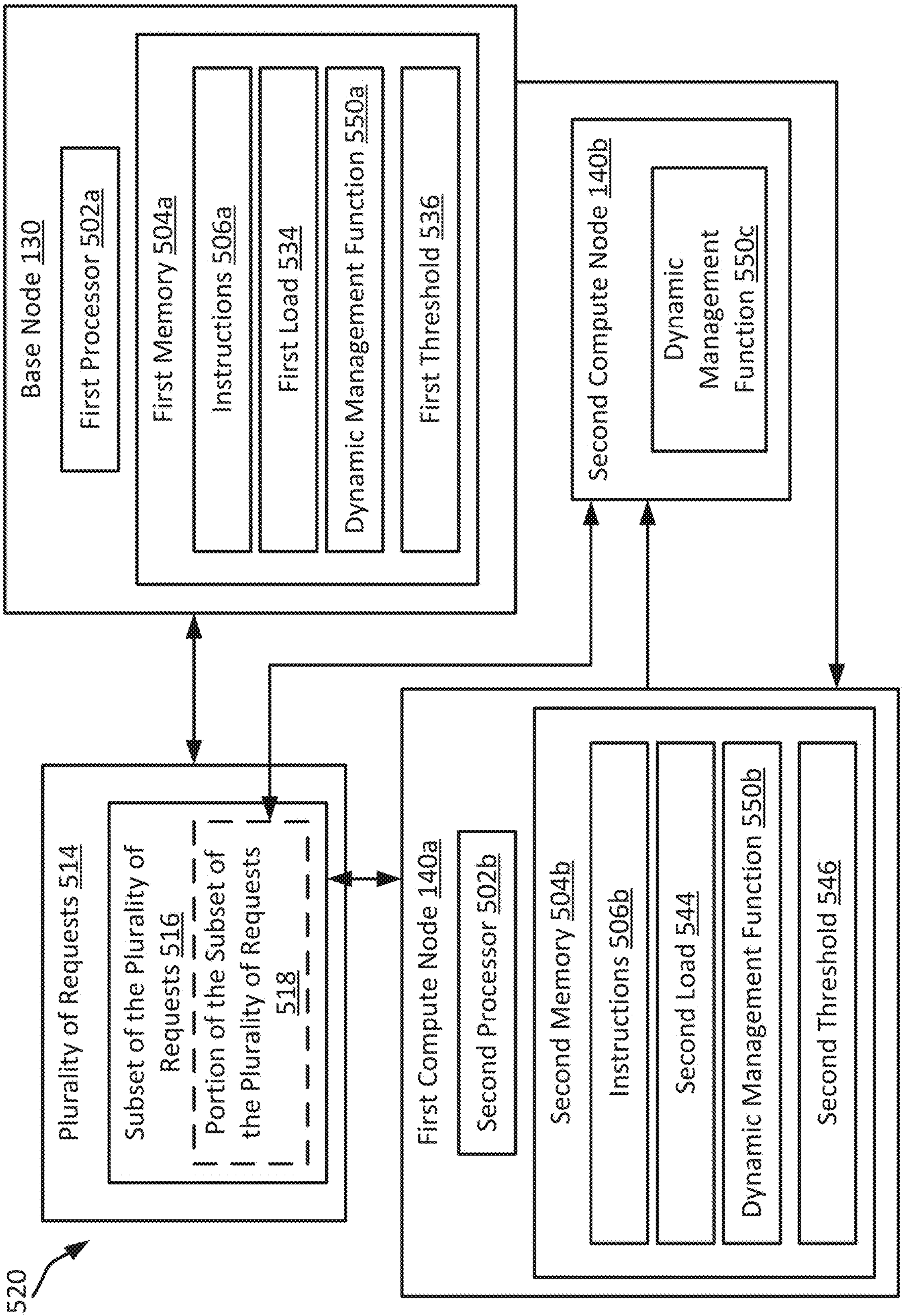

520
Base Node 130
First Processor 502a
First Memory 504a
Instructions 506a
First Load 534
Dynamic Management Function 550a
First Threshold 536
Second Compute Node 140b
Dynamic Management Function 550c
Plurality of Requests 514
Subset of the Plurality of Requests 516
Portion of the Subset of the Plurality of Requests 518
First Compute Node 140a
Second Processor 502b
Second Memory 504b
Instructions 506b
Second Load 544
Dynamic Management Function 550b
Second Threshold 546

FIG. 5

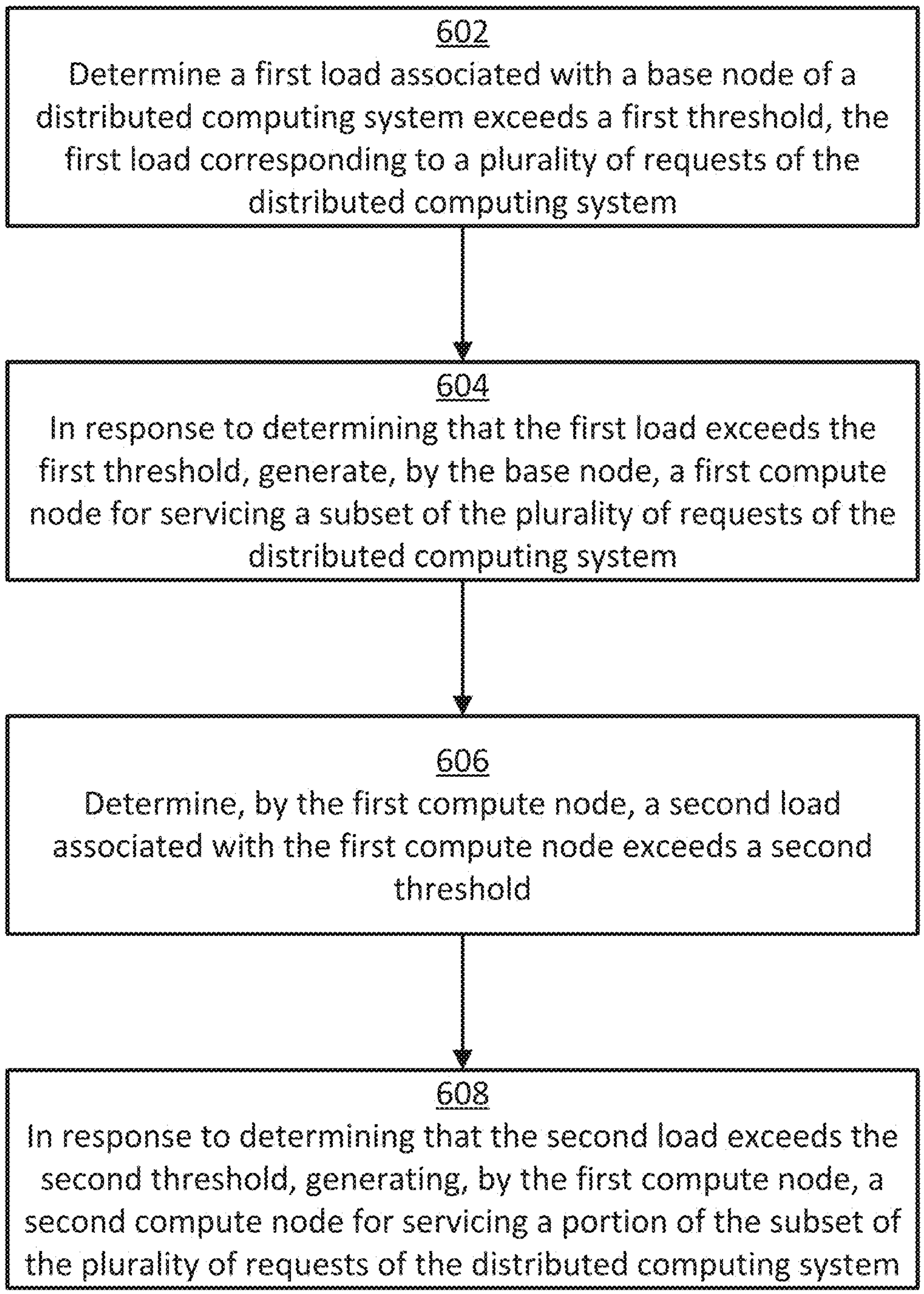

602
Determine a first load associated with a base node of a distributed computing system exceeds a first threshold, the first load corresponding to a plurality of requests of the distributed computing system
604
In response to determining that the first load exceeds the first threshold, generate, by the base node, a first compute node for servicing a subset of the plurality of requests of the distributed computing system
606
Determine, by the first compute node, a second load associated with the first compute node exceeds a second threshold
608
In response to determining that the second load exceeds the second threshold, generating, by the first compute node, a second compute node for servicing a portion of the subset of the plurality of requests of the distributed computing system

FIG. 6

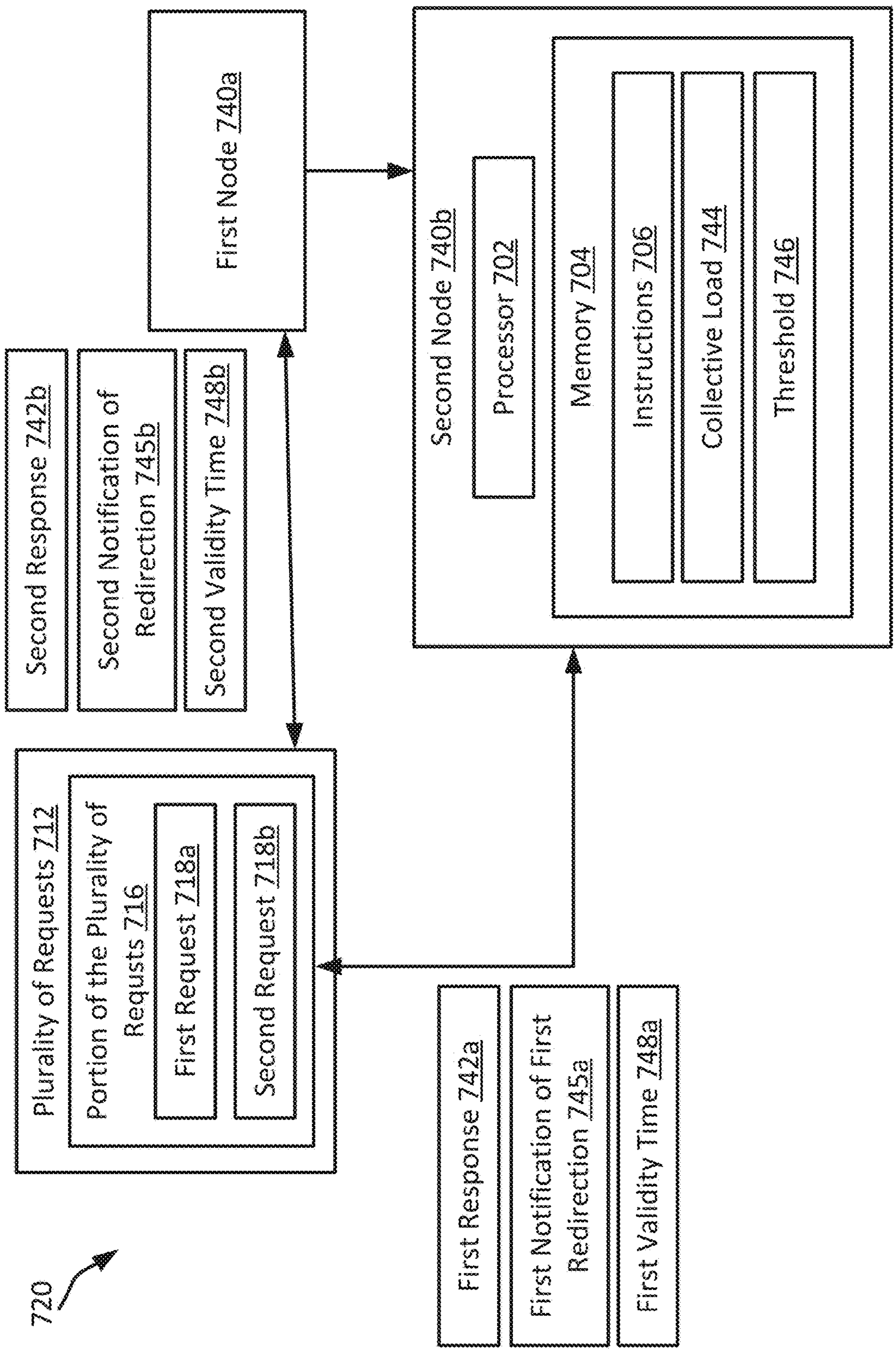

720
First Node 740a
Second Node 740b
Processor 702
Memory 704
Instructions 706
Collective Load 744
Threshold 746
Second Response 742b
Second Notification of Redirection 745b
Second Validity Time 748b
Plurality of Requests 712
Portion of the Plurality of Requsts 716
First Request 718a
Second Request 718b
First Response 742a
First Notification of First Redirection 745a
First Validity Time 748a

FIG. 7

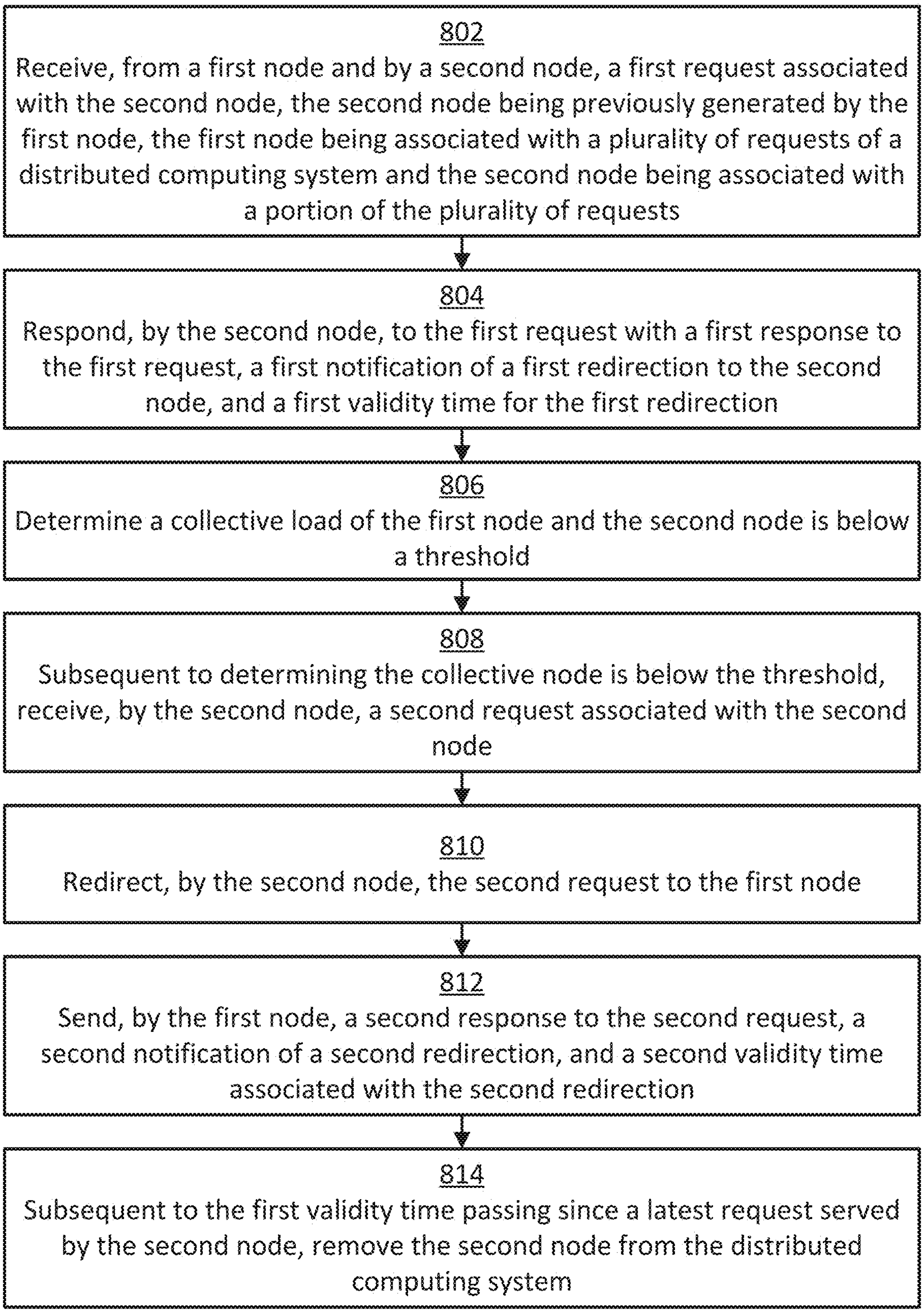

802
Receive, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests
804
Respond, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection
806
Determine a collective load of the first node and the second node is below a threshold
808
Subsequent to determining the collective node is below the threshold, receive, by the second node, a second request associated with the second node
810
Redirect, by the second node, the second request to the first node
812
Send, by the first node, a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection
814
Subsequent to the first validity time passing since a latest request served by the second node, remove the second node from the distributed computing system

FIG. 8

DYNAMIC SCALING OF A DISTRIBUTED COMPUTING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This is a continuation of U.S. patent application Ser. No. 17/530,215, filed Nov. 18, 2021, titled "DYNAMIC SCALING OF A DISTRIBUTED COMPUTING SYSTEM," the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates generally to distributed computing systems. More specifically, but not by way of limitation, this disclosure relates to dynamic scaling of a distributed computing system.

BACKGROUND

[0003] There are various types of distributed computing environments, such as cloud computing systems, computing clusters, and data grids. A distributed computing system can include multiple nodes (e.g., physical machines or virtual machines) in communication with one another over a network, such as a local area network or the Internet. Cloud computing systems have become increasingly popular. Cloud computing environments have a shared pool of computing resources (e.g., servers, storage, and virtual machines) that are used to provide services to developers on demand. These services are generally provided according to a variety of service models, such as Infrastructure as a Service, Platform as a Service, or Software as a Service. But regardless of the service model, cloud providers manage the physical infrastructures of the cloud computing environments to relieve this burden from developers, so that the developers can focus on deploying software applications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. **1** shows a block diagram of an example of a system for implementing dynamic scaling of a distributed computing system according to some aspects of the present disclosure.

[0005] FIG. **2** shows a block diagram of an example of a base node at an initial stage of dynamic scaling of a distributed computing system according to some aspects of the present disclosure.

[0006] FIG. **3** shows a block diagram of an example of nodes after a first expansion of a distributed computing system according to some aspects of the present disclosure.

[0007] FIG. **4** shows a block diagram of an example of nodes after additional expansions of a distributed computing system according to some aspects of the present disclosure.

[0008] FIG. **5** shows a block diagram of an example of a system for implementing dynamic expansion of a distributed computing system according to some aspects of the present disclosure.

[0009] FIG. **6** shows a flow chart of an example of a process for implementing dynamic expansion of a distributed computing system according to some aspects of the present disclosure.

[0010] FIG. **7** shows a block diagram of an example of a system for implementing dynamic reduction of a distributed computing system according to some aspects of the present disclosure.

[0011] FIG. **8** shows a flow chart of an example of a process for implementing dynamic reduction of a distributed computing system according to some aspects of the present disclosure.

DETAILED DESCRIPTION

[0012] Distributed computing systems, such as cloud computing systems and cluster computing systems, can support dynamic expansion and shrinking of compute nodes. A distributed computing system typically involves a scaling management system with a proxy that may peek into the requests payload and performs redirections according to a scaling configuration. The scaling mechanism includes a central component for management and redirection associated with all nodes of the distributed computing system. The central component is a single point of failure and is often a bottleneck of the distributed computing system, since all requests and decisions are managed by the central component. As a result, the central component may be complex and expensive to implement.

[0013] Some examples of the present disclosure can overcome one or more of the abovementioned problems by providing a system with a scalable, self-managed scheme that does not involve a central component for the management and redirection. The system can include a base node for providing an access point to a distributed computing system and for serving a first portion of requests. The base node can generate at least one compute node based on a first load of the base node. The compute node can service a second portion of requests. The compute node can generate an additional compute node for servicing a subset of the second portion of requests based on a second load of the compute node. The compute node can also monitor its loads to determine when the compute node can be removed from the system. Since the base node and each of the compute nodes can monitor their respective loads and generate new nodes, the management and redirection of requests from client applications can be considered to be dynamic and self-managed. The system may add little payload that can enable client applications generating the requests to be updated about the dynamic changes while keeping the system functional for non-updated client applications. Additionally, there is no single point of failure of the distributed computing system, and without the bottleneck of the central component, performance of the distributed computing system can be improved.

[0014] As one example, a system can dynamically expand a number of compute nodes of a distributed computing system. The system can include a base node that provides ten client applications access to the distributed computing system. Each of the client applications can generate requests associated with a customer identifier (ID) associated with the request. The base node can monitor the load of requests received from the client applications. Upon determining that the load exceeds a first threshold, the base node can generate a compute node A for servicing a portion of the requests from the client applications. For example, the compute node A can service requests with an even customer ID (e.g., two, four, six, eight, and ten), while the base node continues to service requests with an odd customer ID (e.g., one, three, five, seven, and nine). The base node and the compute node A can then monitor their respective loads. If the compute node A determines that the load of the compute node A exceeds a second threshold, the compute node A can generate a compute node B. The compute node B can service requests for a portion of the requests serviced by the compute node A. For example, the compute node B can service requests with a customer ID that is divisible by four (e.g., four and eight). The base node, compute node A, and compute node B can continue monitoring their respective loads and generating additional compute nodes. Each additional compute node can also monitor their load and generate additional compute nodes. Since compute nodes can individually determine when to generate additional compute nodes, a central component is not needed for managing the expansion of the number of compute nodes.

[0015] As another example, a system can dynamically reduce a number of compute nodes of a distributed computing system. Node A, which may be a base node or a compute node, may have previously generated node B, which can be a compute node. Node B can determine a collective load of node A and node B is below a limit. As a result, node B can determine that it can unite with node A. Prior to determining to unite with node A, node A can receive a request having a customer ID associated with node B. Node A can forward the request to node B, which can respond with a response to the request, a notification of redirection to node B, and a validity time of five minutes for redirecting requests to node B. Subsequent to determining to unite with node A, node B can receive another request having a customer ID associated with node B. The request can be redirected from node B to node A. Node A can send a response to the request, a notification of the redirection to node A, and a validity time associated with the redirection. Once the initial validity time indicated in the response from node B has passed, node B can remove itself from the distributed computing system. Since compute nodes can individually determine when to be removed from the distributed computing system, a central component is not needed for managing the reduction of the number of compute nodes.

[0016] These illustrative examples are given to introduce the reader to the general subject matter discussed here and are not intended to limit the scope of the disclosed concepts. The following sections describe various additional features and examples with reference to the drawings in which like numerals indicate like elements but, like the illustrative examples, should not be used to limit the present disclosure.

[0017] FIG. **1** shows a block diagram of an example of a system **100** for implementing distributed scaling of a distributed computing system **120** according to some aspects of the present disclosure. The system **100** includes one or more client applications **112** in communication with the distributed computing system **120**. The distributed computing system **120** may be a distributed storage system, a customer relationship management (CRM) system, or a cluster computing system. Each client application may be associated with one or more client devices that can access the distributed computing system **120**. The client applications **112** can generate requests **114***a-c*. Examples of the client devices that the client applications **112** can be associated with are a laptop, desktop, mobile device, a server, or other computing device. The client applications **112** can access the distributed computing system **120** via one or more networks, such as a local area network or the Internet. A base node **130** of the distributed computing system **120** can provide an access point **132** for serving requests of the client applications **112**. The access point **132** can include an Internet Protocol (IP) address and a port to the distributed computing system **120**. The distributed computing system **120** can also include one or more compute nodes **140***a-b*. The compute nodes **140***a-b* can each be in communication with some or all of the client applications **112**, the base node **130**, the other compute nodes, or a combination thereof. Examples of the base node **130** and the compute nodes **140***a-b* include a bare metal server, a virtual server, or a proxy server that redirects requests to a highly available cluster, such as an active-passive cluster.

[0018] The distributed computing system **120** may initially include the base node **130** and neither the compute node **140***a* nor the compute node **140***b*. The base node **130** can service requests **114***a-c* for each of the client applications **112**. The base node **130** can include a dynamic management function for monitoring a load **134** of the base node **130** and generating compute nodes accordingly. For example, the base node **130** may monitor a compute load, a network load, a memory load, a storage load, a combination thereof, etc. to determine when the load **134** exceeds a threshold **136**. Upon determining that the load **134** exceeds the threshold **136**, the base node **130** can generate the compute node **140***a* for servicing a portion of the requests **114***a-c*. For example, the base node **130** may service requests **114***a* and the compute node **140***a* may service requests **114***b-c*. The requests **114***b-c* can include a similar characteristic. For example, the requests **114***a* may be associated with an even customer ID and the requests **114***b-c* may be associated with an odd customer ID. In a CRM system, it may be beneficial to have all requests for a particular customer to be handled by same node. Otherwise, coordination between multiple nodes handling the requests for the customer can be complex and computationally expensive. Associating the requests **114***a-c* based on customer IDs can thus reduce the complexity and computation requirements of handling requests. Additionally or alternatively, the requests **114***a-c* may be distributed between the base node **130** and the compute node **140***a* based on a geographic location of the requests **114***a-c*, a hash value of a username included in the requests **114***a-c*, or any other function of the request input. For example, the requests **114***a* may be associated with Europe and the requests **114***b-c* may be associated with the United States.

[0019] In some examples, subsequent to the base node **130** generating the compute node **140***a*, the base node **130** and the compute node **140***a* can individually monitor their respective loads. For example, while the base node **130** continues to monitor the load **134**, the compute node **140***a* can monitor a load **144***a* associated with the compute node **140***a*. If the base node **130** determines the load **134** again exceeds the threshold **136**, the base node **130** can generate another compute node for servicing requests from a portion of the remaining requests associated with the base node **130**. Additionally, if the compute node **140***a* determines the load **144***a* exceeds a threshold **146***a*, the compute node **140***a* can generate a compute node for servicing a portion of the requests associated with the compute node **140***a*. For example, the compute node **140***a* may be associated with the requests **114***b-c* and may determine the load **144***a* exceeds the threshold **146***a*. The compute node **140***a* can then generate compute node **140***b* for servicing either the requests **114***b* or the requests **114***c*. The compute node **140***b* can then monitor a load **144***b* associated with the compute node **140***b* and generate an additional compute node if the load **144***b* exceeds the threshold **146***b*. Each of the threshold **136** and the thresholds **146***a-b* may be equal or unequal.

[0020] Once the base node **130** generates the compute node **140***a*, the client applications **112** generating requests previously associated with the base node **130** and currently serviced by the compute node **140***a* may lack knowledge that the compute node **140***a* services the requests. For example, prior to the compute node **140***a* being generated, the base node **130** may service the requests **114***b* that are serviced by the compute node **140***a* once the compute node **140***a* is generated. The client applications **112** may not know the compute node **140***a* has been generated, so a client application of the client applications **112** can send a request of the requests **114***b* to the base node **130**. Since the base node **130** generated the compute node **140***a*, the base node **130** can determine the compute node **140***a* services the requests **114***b*. The base node **130** can forward the request to the compute node **140***a*. The compute node **140***a* can then respond to the request with an indication **116** of the requests that are associated with the compute node **140***a*. For example, the indication **116** can indicate to the client applications **112** that the requests **114***b* are associated with the compute node **140***a*. The indication **116** may include customer IDs associated with the requests **114***b* that are associated with the compute node **140***a*, fields and values (e.g., cities and addresses) associated with the requests **114***b* that are associated with the compute node **140***a*, or a rule for determining the requests **114***b* that are associated with the compute node **140***a*. If the compute node **140***a* services the requests **114***b* with an odd customer ID, the rule may be customerID%2==1. Subsequent requests of the requests **114***b* can then be received by the compute node **140***a* without first being redirected by the base node **130**.

[0021] In some examples, the compute nodes **140***a-b* can also determine when the compute nodes **140***a-b* are to be removed from the distributed computing system **120**. A compute node that does not have a child, that is, a compute node without a generated compute node, can decide to unite with its parent node. The parent node is the node that generated the compute node. For example, if the distributed computing system **120** includes the base node **130** and the compute node **140***a* generated by the base node **130**, the compute node **140***a* may decide to unite with the base node **130** at some point in time. The compute node **140***a* can determine whether a collective load of the base node **130** and the compute node **140***a* is below a limit **148**. Additionally or alternatively, the base node **130** may determine whether the collective load is below the limit **148**. The collective load can be a summation of the load **134** of the base node **130** and the load **144***a* of the compute node **140***a*. If the collective load is below the limit **148**, the compute node **140***a* can be removed from the distributed computing system **120**. Otherwise, the compute node **140***a* is to remain in the distributed computing system **120** for servicing a portion of the requests **114***a-c*.

[0022] To remove the compute node **140***a* from the distributed computing system **120** in response to determining that the collective load is below the limit **148**, the compute node **140***a* can indicate a validity time to the client applications **112**. The validity time can be a time length for which the compute node **140***a* services particular requests. For example, if the compute node **140***a* services the requests **114***b-c* and the compute node **140***a* receives a request redirected from the base node **130**, the indication **116** sent to the client applications **112** in response to the request by the compute node **140***a* can include, in addition to the response to the request, a notification of a redirection of the request to the compute node **140***a*, and a validity time associated with the redirection. The notification of the redirection can be a redirection record and the validity time can be a validity period for the redirection record. As one particular example, the validity time may be five minutes. If the compute node **140***a* receives an additional request from the requests **114***b-c* within the validity period of the request but subsequent to determining to unite with the base node **130**, the additional request can be redirected by the compute node **140***a* to the base node **130**, which can respond to the additional request with an additional notification of redirection and validity time for the base node **130**. Once the validity time for the redirection record of the compute node **140***a* has passed, the client applications **112** can no longer send requests to the compute node **140***a*, and instead can send additional requests to the base node **130**, since routing to the compute node **104***a* is no longer valid. Subsequent to the validity time for the compute node **140***a* passing, the compute node **140***a* can be removed from the distributed computing system **120** and the base node **130** can service the requests **114***b-c*. The base node **130** may not have an expiry time and an address of the base node **130** can be part of an invariant configuration of the distributed computing system **120**.

[0023] Removing the compute node **140***b* from the distributed computing system **120** can follow a similar process. The compute node **140***b* can determine a collective load of the loads **144***a-b* is below a limit and redirect requests to the compute node **140***a*. The compute node **140***a* can respond to the requests with the indication of the response to the request, the redirection to the compute node **140***a*, and a validity time associated with the redirection. After a validity time associated with the compute node **140***b* has passed, the compute node **140***b* can be removed from the distributed computing system **120** and the compute node **140***a* can service the requests previously associated with the compute node **140***b*.

[0024] Although FIG. **1** depicts a certain number and arrangement of components, other examples may include more components, fewer components, different components, or a different number of the components that is shown in FIG. **1**. For instance, the distributed computing system **120** can include more base nodes than are shown in FIG. **1**. Additionally, while two compute nodes are shown in FIG. **1**, in other examples the distributed computing system **120** may include more or fewer compute nodes. Each of the compute nodes can include the dynamic management function for generating additional compute nodes and determining when to be removed from the distributed computing system **120**.

[0025] FIG. **2** shows block diagrams of an example of a base node **230** at an initial stage of dynamic scaling of a distributed computing system according to some aspects of the present disclosure. A single server, illustrated as the base node **230***a*, can handle all requests from client applications **212**, as indicated by S_{all}. A system with only the base node **230***a* can be considered an initial state of the system. The base node **230***a* can monitor a load of the base node **230***a*, and when the load exceeds a threshold the base node **230***a* can spawn a new compute node for serving part of the requests from the client applications **212**. The threshold may be based on quality of service (QOS) targets.

[0026] FIG. **3** show block diagrams of an example of nodes after a first expansion of a distributed computing system according to some aspects of the present disclosure. The base node **230** is illustrated as having generated a compute node **240***a* for serving a portion of the requests from the client applications **212**. The base node **230** can be a gateway to all requests and can handle requests with even customer ID numbers, as indicated by $S_{all,ID\%2==0}$. The compute node **240***a* can handle requests with odd customer IDs, as indicated by $S_{ID\%2==1}$. The client applications **212** may not be aware of the change, so a next call can route to the base node **230** even if the customer ID is odd. If the customer ID is odd, base node **230** re-routes the request to the compute node **240***a*. The compute node **240***a* handles the request and the response returns information about the requests that the compute node **240***a* handles. The response information can be in different semantics, such as the customer ID, a range of customer IDs, groups of fields and values or rules. The most efficient response information may be to return the rule of customerID%2==1. Other examples may include a value of customerAddress.City in a group of cities, or anything else. After a single call that is forwarded from the base node **230** to the compute node **240***a*, the client applications **212** have updated information and send subsequent requests to the compute node **240***a* directly.

[0027] FIG. **4** shows block diagrams of an example of nodes after additional expansions of a distributed computing system according to some aspects of the present disclosure. The base node **230** is illustrated as having generated another compute node, illustrated as compute node **240***b*. The compute node **240***a* is also illustrated as having generated compute node **240***c*. The base node **230** generates the compute node **240***b* based on a load of the base node **230** exceeding a first threshold, and the compute node **240***a* generates the compute node **240***a* based on a load of the compute node **240***a* exceeding a second threshold. In this case, if a request with customerID%4==3 is routed to the base node **230**, the request can be routed twice until the request reaches the compute node **240***c*, but the response can provide the client applications **212** with information about which requests to route to the compute node **240***c* on subsequent calls.

[0028] Although FIG. **4** shows a tree with all branches having a same height, the tree can develop in every direction according to the load, which may create an asymmetrical tree. For example one node may handle all the odd customer IDs, and four nodes can handle even customer IDs according to the residue from dividing the ID by eight.

[0029] FIG. **5** shows a block diagram of an example of a system for implementing dynamic expansion of a distributed computing system **520** according to some aspects of the present disclosure. The distributed computing system **520** includes a first processor **502***a* that is communicatively coupled to a first memory **504***a*. In some examples, the first processor **502***a* and the first memory **504***a* can be part of the same computing device, such as a base node **130**. In other examples, the first processor **502***a* and the first memory **504***a* can be distributed from (e.g., remote to) one another. The distributed computing system **520** also includes a second processor **502***b* that is communicatively coupled to a second memory **504***b*. In some examples, the second processor **502***b* and the second memory **504***b* can be part of the same computing device, such as a first compute node **140***a*. In other examples, the second processor **502***b* and the second memory **504***b* can be distributed from (e.g., remote to) one another.

[0030] The first processor **502***a* and the second processor **502***b* can each include one processor or multiple processors. Non-limiting examples of the first processor **502***a* and the second processor **502***b* include a Field-Programmable Gate Array (FPGA), an application-specific integrated circuit (ASIC), or a microprocessor. The first processor **502***a* can execute instructions **506***a* stored in the first memory **504***a* to perform operations. The second processor **502***b* can execute instructions **506***b* stored in the second memory **504***b* to perform operations. The instructions **506***a-b* may include processor-specific instructions generated by a compiler or an interpreter from code written in any suitable computer-programming language, such as C, C++, C#, Java, or Python.

[0031] The first memory **504***a* and the second memory **504***b* can each include one memory or multiple memories. The first memory **504***a* and the second memory **504***b* can be volatile or non-volatile. Non-volatile memory includes any type of memory that retains stored information when powered off. Examples of the first memory **504***a* and the second memory **504***b* include electrically erasable and programmable read-only memory (EEPROM) or flash memory. At least some of the first memory **504***a* can include a non-transitory computer-readable medium from which the first processor **502***a* can read instructions **506***a*. At least some of the second memory **504***b* can include a non-transitory computer-readable medium from which the second processor **502***b* can read instructions **506***b*. A non-transitory computer-readable medium can include electronic, optical, magnetic, or other storage devices capable of providing the first processor **502***a* and the second processor **502***b* with computer-readable instructions or other program code. Examples of a non-transitory computer-readable medium can include a magnetic disk, a memory chip, ROM, random-access memory (RAM), an ASIC, a configured processor, and optical storage.

[0032] The first processor **502***a* can execute the instructions **506***a* to perform operations. For example, the first processor **502***a* can determine a first load **534** associated with the base node **130** of the distributed computing system **520** having a dynamic management function **550***a* exceeds a first threshold **536**. The first load **534** can be a compute load, network load, memory load, storage load, a combination thereof, etc. corresponding to a plurality of requests **514** of the distributed computing system **520**. The plurality of requests **514** can be generated by a plurality of client applications. In response to determining that the first load **534** exceeds the first threshold **536**, the first processor **502***a* can generate, by the base node **130**, the first compute node **140***a* for servicing a subset of the plurality of requests **516** of the distributed computing system **520**. The subset of the plurality of requests **516** can have a similar characteristic, such as a customer ID, geographic location, or username hash value that meets a particular criteria.

[0033] The first compute node **140***a* can have the dynamic management function **550***b*, and the second processor **502***b* can execute the instructions **506***b* to perform operations. For example, the second processor **502***b* can determine, by the first compute node **140***a*, a second load **544** associated with the first compute node **140***a* exceeds a second threshold **546**. In response to determining the second load **544** exceeds the second threshold **546**, the second processor **502***b* can generate, by the first compute node **140***a*, a second compute node **140***b* for servicing a portion of the subset of the plurality of requests **518** of the distributed computing system **520**. The second compute node **140***b* can have the dynamic management function **550***c*. Accordingly, each node of the distributed computing system **520** can monitor their respective loads and generate additional compute nodes to service a portion of their associated requests. Thus, the distributed computing system **520** does not include a central management component that may be a bottleneck for the system when the central management component controls the generation of compute nodes.

[0034] FIG. **6** shows a flow chart of a process for implementing dynamic expansion of a distributed computing system according to some aspects of the present disclosure. Other examples can include more steps, fewer steps, different steps, or a different order of the steps than is shown in FIG. **6**. The steps of FIG. **6** are discussed below with reference to the components discussed above in relation to FIG. **5**.

[0035] In block **602**, a first processor **502***a* can determine a first load **534** associated with a base node **130** of a distributed computing system **520** exceeds a first threshold **536**. The first load **534** can correspond to a plurality of requests **514** of the distributed computing system **520**. The first load **534** can be a compute load, a network load, storage load, memory load, a combination thereof, etc. associated with the base node **130**. The plurality of requests **514** can be associated with a first geographic region, a first portion of customer IDs, or a combination thereof. The base node **130** can provide an access point for client applications that generate the plurality of requests **514** to the distributed computing system **520**.

[0036] In block **604**, in response to determining that the first load **534** exceeds the first threshold **536**, the first processor **502***a* can generate, by the base node **130**, a first compute node **140***a* for servicing a subset of the plurality of requests **516** of the distributed computing system **520**. The subset of the plurality of requests **516** can be associated with a second geographic region within the first geographic region, a second portion of customer IDs (e.g., even customer IDs), or a combination thereof. Once the first compute node **140***a* is generated, a request associated with the first compute node **140***a* may initially be received by the base node **130** and redirected to the first compute node **140***a*. The first compute node **140***a* can respond to the request with an indication of the subset of the plurality of requests **516** that are associated with the first compute node **140***a*.

[0037] In block **606**, a second processor **502***b* of the first compute node **140***a* can determine a second load **544** associated with the first compute node **140***a* exceeds a second threshold **546**. The second load **544** can correspond to the subset of the plurality of requests **516** of the distributed computing system **520**. The second load **544** can be a compute load, a network load, a memory load, a storage load, a combination thereof, etc. associated with the first compute node **140***a*.

[0038] In block **608**, in response to determining the second load **544** exceeds the second threshold **546**, the second processor **502***b* can generate, by the first compute node **140***a*, a second compute node **140***b* for servicing a portion of the subset of the plurality of requests **518** of the distributed computing system **520**. The portion of the subset of the plurality of requests **518** can be associated with a third geographic region within the second geographic region, a second portion of customer IDs within the first portion of customer IDs, or a combination thereof. The second compute node **140***b* can monitor an associated load of the second compute node **140***b* and generate additional compute nodes for further distributing the plurality of requests **514** across the base node **130** and the compute nodes.

[0039] FIG. **7** shows a block diagram of an example of a system for implementing dynamic reduction of a distributed computing system **720** according to some aspects of the present disclosure. The distributed computing system **720** includes a second node **740***b* with a processor **702** communicatively coupled to a memory **704**. The second node **740***b* can be communicatively coupled to a first node **740***a*. The first node **740***a* may be a base node (e.g., base node **130** in FIGS. **1** and **5**) or a first compute node (e.g., first compute node **140***a* in FIGS. **1** and **5**). If the first node **740***a* is the base node, the second node **740***b* may be the first compute node. If the first node **740***a* is the first compute node, the second node **740***b* can be a second compute node (e.g., second compute node **140***b* in FIGS. **1** and **5**).

[0040] The processor **702** can execute instructions **706** stored in memory **704** to perform operations, such as any of the operations described herein. If the second node **740***b* is the first compute node, the instructions **706** can be the instructions **506***b* in FIG. **5**. For example, the processor **702** can receive, from the first node **740***a* and by the second node **740***b*, a first request **718***a* associated with the second node **740***b*. The second node **740***b* can be previously generated by the first node **740***a*. The first node **740***a* can be associated with a plurality of requests **712** of the distributed computing system **720** and the second node **740***b* can be associated with a portion of the plurality of requests **716**. The second node **740***b* can respond to the first request **718***a* with a first response **742***a* to the first request **718***a*, a first notification of a first redirection **745***a* to the second node **740***b*, and a first validity time **748***a* for the first redirection. The second node **740***b* can determine a collective load **744** of the first node **740***a* and the second node **740***b* is below a threshold **746**. Subsequent to determining the collective load **744** is below the threshold **746**, the second node **740***b* can receive a second request **718***b* associated with the second node **740***b*. The second node **740***b* can redirect the second request **718***b* to the first node **740***a*. The first node **740***a* can then send a second response **742***b* to the second request **718***b*, a second notification of a second redirection **745***b*, and a second validity time **748***b* associated with the second redirection. Subsequent to the first validity time **748***a* passing since a latest request served by the second node **740***b*, the second node **740***b* can be removed from the distributed computing system **720**. Accordingly, each node of the distributed computing system **720** can monitor their respective loads and remove themselves from the distributed computing system **720** to minimize resources used by the distributed computing system **720**.

[0041] FIG. **8** shows a flow chart of an example of a process for implementing dynamic reduction of a distributed computing system according to some aspects of the present disclosure. Other examples can include more steps, fewer steps, different steps, or a different order of the steps than is shown in FIG. **8**. The steps of FIG. **8** are discussed below with reference to the components discussed above in relation to FIG. **7**.

[0042] In block **802**, a processor **702** can receive, from a first node **740***a* and by a second node **740***b*, a first request **718***a* associated with the second node **740***b*. The second node **740***b* can be previously generated by the first node **740***a*. The first node **740***a* may correspond to a base node (e.g., base node **130** in FIGS. **1** and **5**) or a first compute node (e.g., first compute node **140***a* in FIGS. **1** and **5**). If the first node **740***a* is the base node, the second node **740***b* can correspond to the first compute node. If the first node **740***a* is the first compute node, the second node **740***b* can correspond to a second compute node (e.g., second compute node **140***b* in FIGS. **1** and **5**). The first node **740***a* can be associated with a plurality of requests **712** from client applications of a distributed computing system **720** and the second node **740***b* can be associated with a portion of the plurality of requests **716**. The plurality of requests **712** can be associated with a first geographic region, a first portion of customer IDs, or a combination thereof, and the portion of the plurality of requests **716** can be associated with a second geographic region within the first geographic region, a second portion of customer IDs (e.g., even customer IDs), or a combination thereof.

[0043] In block **804**, the processor **702** can respond, by the second node **740***b*, to the first request **718***a* with a first response **742***a* to the first request **718***a*, a first notification of a first redirection **745***a* to the second node **740***b*, and a first validity time **748***a* for the first redirection. For example, the first validity time **748***a* may be three minutes, so requests associated with the second node **740***b* within the first validity time **748***b* can be automatically sent to the second node **740***b* without first being sent to and redirected by the first node **740***a*.

[0044] In block **806**, the processor **702** can determine a collective load **744** of the first node **740***a* and the second node **740***b* is below a threshold **746**. The first node **740***a* or the second node **740***b* may determine that the collective load **744** is below the threshold **746**.

[0045] In block **808**, subsequent to determining the collective load **744** is below the threshold **746**, the processor **702** can receive a second request **718***b* associated with the second node **740***b*. The client application sending the second request **718***b* may have previously been notified that the second node **740***b* is associated with the second request **718***b*.

[0046] In block **810**, the processor **702** can redirect the second request **718***b* to the first node **740***a*. The second request **718***b* can be redirected after the first node **740***a* and the second node **740***b* decide collectively to remove the second node **740***b* from the distributed computing system **720** since the collective load **744** is below the threshold **746**.

[0047] In block **812**, the first node **740***a* can send a second response **742***b* to the second request **718***b*, a second notification of a second redirection **745***b*, and a second validity time **748***b* associated with the second redirection. The second notification of the second redirection **745***b* can indicate that the first node **740***a* is associated with the second request **718***b*. The second validity time **748***b* can be a time associated with an expiration of the second redirection. For example, the second validity time **748***b* may be one minute, indicating that after one minute has passed, the first node **740***a* may no longer be associated with the second request **718***b*.

[0048] In block **814**, subsequent to the first validity time **748***a* passing since a latest request served by the second node **740***b*, the processor **702** can remove the second node **740***b* from the distributed computing system **720**. Once the first validity time **748***a* passes, any redirection to the second node **740***b* indicated to the client applications for the portion of the plurality of requests **716** is expired, so the client applications send requests to a base node, which may be the first node **740***a*. As a result, the client applications will not attempt to send a request to the second node **740***b* subsequent to the second node **740***b* being removed from the distributed computing system.

[0049] As used below, any reference to a series of examples is to be understood as a reference to each of those examples disjunctively (e.g., "Examples 1-4" is to be understood as "Examples 1, 2, 3, or 4").

[0050] Example 1 is a system comprising: a base node configured to provide an access point to a distributed computing system and for servicing a first portion of requests and to generate at least one compute node based on a first load of the base node, the at least one compute node configured to service a second portion of requests and to generate an additional compute node for servicing a subset of the second portion of requests based on a second load of the at least one compute node.

[0051] Example 2 is the system of example(s) 1, wherein the base node is further configured to: receive, subsequent to generating the at least one compute node, a request associated with the at least one compute node; send the request to the at least one compute node; and wherein the at least one compute node is further configured to: receive the request from the base node; and respond to the request with an indication of the second portion of requests associated with the at least one compute node.

[0052] Example 3 is the system of example(s) 2, wherein the indication of the second portion of requests includes customer identifiers associated with the second portion of requests, fields and values associated with the second portion of requests, or a rule for determining requests associated with the second portion of requests.

[0053] Example 4 is the system of example(s) 1, wherein the base node is configured to generate the at least one compute node for servicing the second portion of requests by: determining the first load associated with the base node exceeds a first threshold, the first load corresponding to the first portion of requests; and in response to determining that the first load exceeds the first threshold, generating, by the base node, the at least one compute node for servicing the second portion of requests.

[0054] Example 5 is the system of example(s) 4, wherein the at least one compute node is configured to generate the additional compute node for servicing the subset of the second portion of requests by: determining, by the at least one compute node, the second load associated with the at least one compute node exceeds a second threshold; and in response to determining the second load exceeds the second threshold, generating, by the at least one compute node, the additional compute node for servicing the subset of the second portion of requests.

[0055] Example 6 is the system of example(s) 2, wherein the indication of the second portion of requests comprises a first notification of a first redirection to the at least one compute node and a first validity time for the first redirection, and the at least one compute node is further configured to: determine a collective load of the base node and the at least one compute node is below a limit; subsequent to determining the collective load is below the limit, receive, by the at least one compute node, an additional request associated with the at least one compute node; redirect, by the at least one compute node, the additional request to the base node; send, by the base node, a response to the additional request, a second notification of a second redirection of the additional request, and a second validity time associated with the second redirection; and subsequent to the first validity time passing, shut down the at least one compute node.

[0056] Example 7 is the system of example(s) 1, wherein the additional compute node is configured to generate another compute node for servicing a portion of the subset of the second portion of requests.

[0057] Example 8 is a method comprising: determining a first load associated with a base node of a distributed computing system exceeds a first threshold, the first load corresponding to a plurality of requests of the distributed computing system; in response to determining that the first load exceeds the first threshold, generating, by the base node, a first compute node for servicing a subset of the plurality of requests of the distributed computing system; determining, by the first compute node, a second load associated with the first compute node exceeds a second threshold; and in response to determining that the second load exceeds the second threshold, generating, by the first compute node, a second compute node for servicing a portion of the subset of the plurality of requests of the distributed computing system.

[0058] Example 9 is the method of example(s) 8, further comprising: receiving, from the base node by the first compute node, a request associated with the first compute node; and responding to the request with an indication of the subset of the plurality of requests associated with the first compute node.

[0059] Example 10 is the method of example(s) 9, wherein the indication of the subset of the plurality of requests includes customer identifiers associated with the subset of the plurality of requests, fields and values associated with the subset of the plurality of requests, or a rule for determining requests associated with the subset of the plurality of requests.

[0060] Example 11 is the method of example(s) 9, wherein the indication of the subset of the plurality of requests associated with the first compute node comprises a first notification of a first redirection to the first compute node and a first validity time for the first redirection, the method further comprising: determining a collective load of the first load and the second load is below a limit; subsequent to determining the collective load is below the limit, receiving, at the first compute node, an additional request associated with the first compute node; redirecting, by the first compute node, the additional request to the base node; sending, by the base node, a response to the additional request, a second notification of a second redirection of the additional request, and a second validity time associated with the redirection; and subsequent to the first validity time passing since a latest request served by the first compute node, shut down the first compute node.

[0061] Example 12 is the method of example(s) 8, wherein the base node is configured to provide an access point to the distributed computing system.

[0062] Example 13 is the method of example(s) 8, wherein the plurality of requests corresponds to a first plurality of requests associated with a first geographic region, the subset of the plurality of requests corresponds a second plurality of requests associated with a second geographic region within the first geographic region, and the portion of the subset of the plurality of requests corresponds to a third plurality of requests associated with a third geographic region within the second geographic region.

[0063] 14. A non-transitory computer-readable medium comprising first program code executable by a first processor for causing the first processor to: determine a first load associated with a base node of a distributed computing system having a dynamic management function exceeds a first threshold, the first load corresponding to a plurality of requests of the distributed computing system; in response to determining that the first load exceeds the first threshold, generate, by the base node, a first compute node for servicing a subset of the plurality of requests of the distributed computing system, the first compute node having the dynamic management function and second program code executable by a second processor for causing the second processor to: determine, by the first compute node, a second load associated with the first compute node exceeds a second threshold; and in response to determining the second load exceeds the second threshold, generate, by the first compute node, a second compute node for servicing a portion of the subset of the plurality of requests of the distributed computing system, the second compute node having the dynamic management function.

[0064] Example 15 is the non-transitory computer-readable medium of example(s) 14, wherein the second program code is further executable by the second processor for causing the second processor to: receive, from the base node and at the first compute node, a request associated with the first compute node; and respond to the request, by the first compute node, with an indication of the subset of the plurality of requests associated with the first compute node.

[0065] Example 16 is the non-transitory computer-readable medium of example(s) 15, wherein the indication of the subset of the plurality of requests includes customer identifiers associated with the subset of the plurality of requests, fields and values associated with the subset of the plurality of requests, or a rule for determining requests associated with the subset of the plurality of requests.

[0066] Example 17 is the non-transitory computer-readable medium of example(s) 15, wherein the indication of the subset of the plurality of requests associated with the first compute node comprises a first notification of a first redirection to the first compute node and a first validity time for the first redirection, and the second program code is further executable by the second processor for causing the second processor to: determine a collective load of the first load and the second load is below a limit; subsequent to determining the collective load is below the limit, receive, at the first compute node, an additional request associated with the first compute node; redirect, by the first compute node, the additional request to the base node; send, by the base node, a response to the additional request, a second notification of a second redirection of the additional request, and a second validity time associated with the second redirection; and subsequent to the first validity time passing since a latest request served by the first compute node, shut down the first compute node.

[0067] Example 18 is the non-transitory computer-readable medium of example(s) 14, wherein the base node is configured to provide an access point to the distributed computing system.

[0068] Example 19 is the non-transitory computer-readable medium of example(s) 14, wherein the plurality of requests corresponds to a first plurality of requests associated with a first geographic region, the subset of the plurality of requests corresponds a second plurality of requests associated with a second geographic region within the first geographic region, and the portion of the subset of the plurality of requests corresponds to a third plurality of requests associated with a third geographic region within the second geographic region.

[0069] Example 20 is the non-transitory computer-readable medium of example(s) 14, wherein the first threshold and the second threshold are equal.

[0070] Example 21 is a method comprising: receiving, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests; responding, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection; determining a collective load of the first node and the second node is below a threshold; subsequent to determining the collective load is below the threshold, receiving, by the second node, a second request associated with the second node; redirecting, by the second node, the second request to the first node; sending, by the first node, a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection; and subsequent to the first validity time passing since a latest request served by the second node, removing the second node from the distributed computing system.

[0071] Example 22 is the method of example(s) 21, wherein generating the second node comprises: determining a first load associated with the first node exceeds a first threshold, the first load corresponding to the plurality of requests; and in response to determining that the first load exceeds the first threshold, generating, by the first node, the second node for servicing the portion of the plurality of requests.

[0072] Example 23 is the method of example(s) 21, wherein the first node comprises a base node configured to provide an access point to the distributed computing system.

[0073] Example 24 is the method of example(s) 21, further comprising: receiving, by the second node, the first request associated with the second node from the first node; and responding to the first request with an indication of the portion of the plurality of requests associated with the second node.

[0074] Example 25 is the method of example(s) 24, wherein the indication includes the first response to the first request, the first notification of the first redirection of the first request, and the first validity time associated with the first redirection.

[0075] Example 26 is a system comprising: a processor; and a memory including instructions that are executable by the processor for causing the processor to: receive, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests; respond, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection; determining a collective load of the first node and the second node is below a threshold; subsequent to determining the collective load is below the threshold, receiving, by the second node, a second request associated with the second node; redirecting, by the second node, the second request to the first node; sending, by the first node, a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection; and subsequent to the first validity time passing since a latest request served by the second node, removing the second node from the distributed computing system.

[0076] Example 27 is the system of example(s) 26, wherein the memory further includes instructions that are executable by the processor for causing the processor to generate the second node by: determining a first load associated with the first node exceeds a first threshold, the first load corresponding to the plurality of requests; and in response to determining that the first load exceeds the first threshold, generating, by the first node, the second node for servicing the portion of the plurality of requests.

[0077] Example 28 is the system of example(s) 26, wherein the first node comprises a base node configured to provide an access point to the distributed computing system.

[0078] Example 29 is the system of example(s) 26, wherein the memory further includes instructions that are executable by the processor for causing the processor to: receive, by the second node, the first request associated with the second node from the first node; and respond to the first request with an indication of the portion of the plurality of requests associated with the second node.

[0079] Example 30 is the system of example(s) 29, wherein the indication includes the first response to the first request, the first notification of the first redirection of the first request, and the first validity time associated with the first redirection.

[0080] Example 31 is a system comprising: first processing means for providing an access point to a distributed computing system and for servicing a plurality of requests and for generating second processing means based on a first load of the first processing means; and the second processing means of the distributed computing system for servicing at least a subset of the plurality of requests, the second processing means configured to generate an additional processing means for servicing a subgroup of the subset of the plurality of requests based on a second load of the second processing means.

[0081] The foregoing description of certain examples, including illustrated examples, has been presented only for the purpose of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Numerous modifications, adaptations, and uses thereof will be apparent to those skilled in the art without departing from the scope of the disclosure. For instance, any examples described herein can be combined with any other examples to yield further examples.

1. A method comprising:

receiving, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests;

responding, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection;

determining a collective load of the first node and the second node is below a threshold;

subsequent to determining the collective load is below the threshold, receiving, by the second node, a second request associated with the second node;

redirecting, by the second node, the second request to the first node;

sending, by the first node, a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection; and

subsequent to the first validity time passing since a latest request served by the second node, removing the second node from the distributed computing system.

2. The method of claim **1**, wherein generating the second node comprises:

determining a first load associated with the first node exceeds a first threshold, the first load corresponding to the plurality of requests; and

in response to determining that the first load exceeds the first threshold, generating, by the first node, the second node for servicing the portion of the plurality of requests.

3. The method of claim **1**, wherein the first node comprises a base node configured to provide an access point to the distributed computing system.

4. The method of claim **1**, further comprising:

receiving, by the second node, the first request associated with the second node from the first node; and

responding to the first request with an indication of the portion of the plurality of requests associated with the second node.

5. The method of claim **4**, wherein the indication includes the first response to the first request, the first notification of the first redirection of the first request, and the first validity time associated with the first redirection.

6. The method of claim **1**, wherein the portion of the plurality of requests have a shared characteristic.

7. The method of claim **1**, wherein the plurality of requests are associated with a first geographic region and the portion of the plurality of requests are associated with a second geographic region within the first geographic region.

8. A system comprising:

a first processor; and

a first memory including instructions that are executable by the first processor for causing the first processor to:

receive, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests;

respond, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection;

determine a collective load of the first node and the second node is below a threshold;

subsequent to determining the collective load is below the threshold, receive, by the second node, a second request associated with the second node;

redirect, by the second node, the second request to the first node for causing the first node to send a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection; and

subsequent to the first validity time passing since a latest request served by the second node, remove the second node from the distributed computing system.

9. The system of claim **8**, further comprising:

a second processor; and

a second memory including instructions that are executable by the second processor for causing the second processor to:

generate the second node by:

determining a first load associated with the first node exceeds a first threshold, the first load corresponding to the plurality of requests; and

in response to determining that the first load exceeds the first threshold, generating, by the first node, the second node for servicing the portion of the plurality of requests.

10. The system of claim **8**, wherein the first node comprises a base node configured to provide an access point to the distributed computing system.

11. The system of claim **8**, wherein the first memory further includes instructions that are executable by the first processor for causing the first processor to:

receive, by the second node, the first request associated with the second node from the first node; and

respond to the first request with an indication of the portion of the plurality of requests associated with the second node.

12. The system of claim **11**, wherein the indication includes the first response to the first request, the first notification of the first redirection of the first request, and the first validity time associated with the first redirection.

13. The system of claim **8**, wherein the portion of the plurality of requests have a shared characteristic.

14. The system of claim **8**, wherein the plurality of requests are associated with a first geographic region and the portion of the plurality of requests are associated with a second geographic region within the first geographic region.

15. A non-transitory computer-readable medium comprising first program code executable by a first processor for causing the first processor to:

receive, from a first node and by a second node, a first request associated with the second node, the second node being previously generated by the first node, the first node being associated with a plurality of requests of a distributed computing system and the second node being associated with a portion of the plurality of requests;

respond, by the second node, to the first request with a first response to the first request, a first notification of a first redirection to the second node, and a first validity time for the first redirection;

determine a collective load of the first node and the second node is below a threshold;

subsequent to determining the collective load is below the threshold, receive, by the second node, a second request associated with the second node;

redirect, by the second node, the second request to the first node for causing the first node to send a second response to the second request, a second notification of a second redirection, and a second validity time associated with the second redirection; and

subsequent to the first validity time passing since a latest request served by the second node, remove the second node from the distributed computing system.

16. The non-transitory computer-readable medium of claim **15**, further comprising second program code executable by a second processor for causing the second processor to:

generate the second node by:

determining a first load associated with the first node exceeds a first threshold, the first load corresponding to the plurality of requests; and

in response to determining that the first load exceeds the first threshold, generating, by the first node, the second node for servicing the portion of the plurality of requests.

17. The non-transitory computer-readable medium of claim **15**, wherein the first node comprises a base node configured to provide an access point to the distributed computing system.

18. The non-transitory computer-readable medium of claim **15**, wherein the first program code is further executable by the first processor for causing the first processor to:

receive, by the second node, the first request associated with the second node from the first node; and

respond to the first request with an indication of the portion of the plurality of requests associated with the second node.

19. The non-transitory computer-readable medium of claim **18**, wherein the indication includes the first response to the first request, the first notification of the first redirection of the first request, and the first validity time associated with the first redirection.

20. The non-transitory computer-readable medium of claim **15**, wherein the plurality of requests are associated with a first geographic region and the portion of the plurality of requests are associated with a second geographic region within the first geographic region.

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